Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 25-01-2026**
Time: 09.00Am to 12.00Pm **GTM-19** **Max. Marks: 300**

KEY SHEET

MATHEMATICS

1	**4**	2	**3**	3	**1**	4	**1**	5	**3**
6	**4**	7	**2**	8	**4**	9	**1**	10	**4**
11	**1**	12	**3**	13	**1**	14	**4**	15	**3**
16	**1**	17	**1**	18	**3**	19	**3**	20	**4**
21	**5**	22	**21**	23	**25**	24	**5**	25	**2**

PHYSICS

26	**2**	27	**3**	28	**4**	29	**1**	30	**4**
31	**1**	32	**1**	33	**3**	34	**2**	35	**1**
36	**4**	37	**3**	38	**3**	39	**1**	40	**3**
41	**3**	42	**1**	43	**2**	44	**2**	45	**1**
46	**6**	47	**15**	48	**314**	49	**2**	50	**150**

CHEMISTRY

51	**1**	52	**1**	53	**1**	54	**3**	55	**3**
56	**1**	57	**2**	58	**3**	59	**2**	60	**3**
61	**2**	62	**4**	63	**2**	64	**4**	65	**2**
66	**1**	67	**2**	68	**2**	69	**1**	70	**2**
71	**6**	72	**6**	73	**5**	74	**6**	75	**56**

For More Join: @IITJEE_Advanced

SOLUTION

MATHEMATICS

1 Put $x-4=\frac{1}{t}\Rightarrow dx=\frac{1}{t^2}dt$

$$I=\int\frac{-\frac{1}{t^2}dt}{\left(\frac{1}{t}\right)^{3/2}\left(-1-\frac{1}{t}\right)^{3/2}}=-\int\frac{dt}{\sqrt{-t-1}}=\frac{-2\sqrt{-t-1}}{-1}+c$$

$$=2\sqrt{\frac{1}{4-x}-1}+c=2\sqrt{\frac{x-3}{4-x}}+c$$

2. $|A|=5$ $\qquad\frac{|adj(adjA)|}{|5A|}=\frac{|A|^4}{5^3|A|}=\frac{5^4}{5^4}=1$

3. $\frac{dx}{dy}-x=y+1$

Solution $\Rightarrow xe^{-y}\int(y+1)e^{-y}dy \qquad \Rightarrow-(y+1)e^{-y}-e^{-y}+c \Rightarrow x+y+2=c.e^{y}$

4. Arranging in alphabetical order $\rightarrow$ C,C,H,I,N,O

CC.... $\rightarrow 4!$
CH.... $\rightarrow 4!$
CI.... $\rightarrow 4!$
CN.... $\rightarrow 4!$
COCHIN.... $\rightarrow 1$
$\therefore$ Number of words before COCHIN $=4!+4!+4!+4!=96$

5. $\begin{vmatrix}\lambda & \lambda+1 & \lambda-1\\ \lambda+1 & \lambda & \lambda+2\\ \lambda-1 & \lambda+2 & \lambda\end{vmatrix}=0$

$C_2\rightarrow C_2-C_1$

$C_3\rightarrow C_3-C_1 \qquad \Rightarrow\begin{vmatrix}\lambda & \lambda+1 & \lambda-1\\ 1 & -1 & 3\\ -1 & 1 & 1\end{vmatrix}=0$

$\Rightarrow\lambda(-1-3)-(\lambda+1)(1+3)+(\lambda-1)(1-1)=0$

$\Rightarrow -4\lambda-4\lambda-4=0 \qquad \Rightarrow 8\lambda+4=0\Rightarrow\lambda=\frac{-1}{2}$

6. Put x = 1 ; f (1) = -1

$x=2$ & $\frac{1}{2}$ $\qquad f(2)+f\left(\frac{1}{2}\right)=1$ _____(2)

$$\&\ 2.f\left(\frac{1}{2}\right)+\frac{1}{2}.f(2)=\frac{1}{2}\text{ _____(3)}$$

For More Join: @IITJEE_Advanced

Solving $f(2)=1, f\left(\frac{1}{2}\right)=0$

7. Let a general point on line L is $(2t+2,3-3t,t-1)$. DRs of AP $2t-1,4-3t,t-5$ for this point to be point P, as AP $\perp$ line L

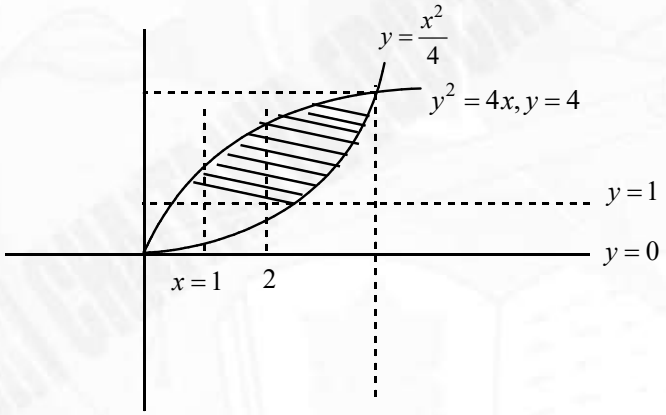
$$2(2t-1)-3(4-3t)+(t-5)=0 \Rightarrow t=\frac{19}{14} \Rightarrow P\left(\frac{33}{7},\frac{-15}{14},\frac{5}{14}\right)$$

8. The numbers upto 25 which gives g.c.d with 24 equals to 1 are 1, 5, 7,11,13,17,19 and 23.
Sum of these numbers = 96
There are four such blocks and a number 97 is there upto 100.
$\therefore$ Complete sum
$= 96+(24\times 8+96)+(48\times 8+96)+(72\times 8+96)+97$
$= 1633$

9. CONCEPTUAL

10. The two curves intersect at (4, 4) which in avertex of the given square.
$\therefore$ Required Area (shaded)

$$\int_1^4 2\sqrt{x}-\int_2^4 \frac{x^2}{4}dx - (2-1)\times 1 \qquad =\frac{11}{3}$$



11. $\overline{AB}+\overline{BC}+\overline{CA}=\bar{0}$
$(-2-\alpha+4)\bar{i}+(1-\beta+3)\bar{j}+(3-\gamma+\delta)\bar{k}=\bar{0}$
$\alpha=2 \quad \beta=4 \quad \gamma-\delta=3$
$\frac{1}{2}\left|\overline{AB}\times\overline{AC}\right|=5\sqrt{6}$
$(9-\delta)^2+(2\delta+12)^2+100=600$
$\therefore \delta=5 \quad \therefore \gamma=8$
$\overline{CB}.\ \overline{CA}=60$

12. Given relation $R=\{(a,b):1+ab>0\}$
Reflexive A relation $R\subseteq A\times A$ is reflexive, if $(a,a)\in R$
Now, $R=\{(a,b)1+ab>0\}$ on Z.
$\Rightarrow (a,a)\in R \Rightarrow 1+a^2>0$

For More Join: @IITJEE_Advanced

So, it is reflexive.

Symmetric if $(a,a)\in R \Rightarrow (b,a)\in R$

Since, $1+ab>0 \Rightarrow 1+ba>0 \quad \Rightarrow (b,a)\in R$

So, it is symmetric.

Transitive if $(a,b)\in R$ *and* $(b,c)\in R \Rightarrow (a,c)\in R$

Now, $(a,b)\in R$ *and* $(b,c)\in R$

$\Rightarrow 1+ab>0$ *and* $1+bc>0 \quad \Rightarrow 2+b(a+c)>0$

i.e., $(a,c)\notin R$

So, it is not transitive.

13. Use $\cos 2A = 2\cos^2 A - 1$

14. $P_{n+2} - P_{n+1} - 7P_n = 0$

$n = 18 \Rightarrow P_{20} - P_{19} - 7P_{18} = 0$

$n = 19 \Rightarrow P_{21} - P_{20} - 7P_{19} = 0$

$(P_{20} - P_{19})(P_{21} - P_{20}) = 49\, P_{18}P_{19}$

15. $\mu = 5.5$

$\sum x_i = 55$

Correct sum

$\sum x_i = 55 - 4 - 5 + 6 + 8 = 60$

Correct mean = 6

Correct $\sum x_i^2 = 371 - 16 - 25 + 36 + 64 = 471 - 41$

$\text{Variance} = \frac{430}{10} - 36 = 7$

16. $\frac{x^2}{4} + \frac{y^2}{2} = 1$ _____(1) $\quad y = x + 1$ ___(2)

eliminating y $3x^2 + 4x - 2 = 0$

x_1, x_2 are the roots $\therefore |x_1 - x_2| = \frac{\sqrt{40}}{3} = |y_1 - y_2|$

$\therefore PQ = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{\frac{40}{9} + \frac{40}{9}} \therefore r = \frac{\sqrt{20}}{3}$

17. $\underset{x\to\frac{\pi}{2}^-}{Zt} f(x) = \underset{x\to\frac{\pi}{2}^-}{Lt}\left(\frac{4}{3}\right)^{\frac{\tan 4x}{\tan 3x}} = \underset{h\to 0}{Lt}\left(\frac{4}{3}\right)^{\frac{\tan 4\left(\frac{\pi}{2}-h\right)}{\tan 3\left(\frac{\pi}{2}-h\right)}} = \underset{h\to 0}{\text{Lt}}\left(\frac{4}{3}\right)^{\frac{\tan 4h}{\cot 3h}}$

$= \underset{h\to 0}{\text{Lt}}\left(\frac{4}{3}\right)^{\frac{\frac{\tan 4h}{h}}{\frac{\cot 3h}{h}}} = 1$

For More Join: @IITJEE_Advanced

$= \underset{x\to\frac{\pi+}{2}}{Lt} f(x) = \underset{x\to\frac{\pi}{2}}{Lt}\left(1+|\cos x|\right)^{\frac{a}{b}|\tan x|}$

$= \underset{x\to\frac{\pi}{2}}{Lt}|1-\cos x|^{\frac{-a}{b}\tan x} = \underset{h\to 0}{Lt}(1+\sinh)^{\frac{+a}{b}\coth}$

$= e^{\underset{h\to 0}{Lt}(1+\sinh-1)\frac{+a}{b}\frac{\cosh}{\sinh}} \therefore e^{\frac{a}{b}} \therefore 1 = b+2 = e^{\frac{a}{b}}$

$b=-1 \quad a=0 \quad \therefore a+b=-1$

18. $\left[\frac{100}{5}\right]+\left[\frac{100}{25}\right] = 20+4 = 24$

19. $T_6+T_{19} = T_{10}+T_{15} = T_{12}+T_{13} = x$

$3x = 600$

$x = 200 = T_1+T_{24}$

$S_{24} = \frac{24}{2}(T_1+T_{24}) = 2400$

20. Use L Hospital rule

21. $\left(\sqrt{a^2e^2+b^2}\right)^2+\left(\sqrt{a^2e^2+b^2}\right)^2 = (2ae)^2$

$2.(a^2e^2+b^2) = 4a^2e^2 \Rightarrow e^2 = \frac{b^2}{a^2}.....(1)$

$\therefore e^2 = \frac{1-b^2}{a^2} = 1-e^2 \qquad \therefore e^2 = \frac{1-b^2}{a^2} = 1-e^2$

$e = \frac{1}{\sqrt{2}} \Rightarrow \frac{\alpha}{\sqrt{\beta}} = e \qquad \therefore \text{Then } \alpha+\beta^2 = 1+2^2 = 5$

22. Let $\alpha = \int_{\frac{1}{2}}^{2} \frac{\tan^{-1}x}{2x^2-3x+2}dx$

put $x = \frac{1}{t}$ $\alpha = \int_{2}^{\frac{1}{2}} \frac{\tan^{-1}\left(\frac{1}{t}\right)}{2\left(\frac{1}{t}\right)^2-3\left(\frac{1}{t}\right)+2}.\left(\frac{-1}{t^2}\right)dt$

$\alpha = \int_{\frac{1}{2}}^{2} \frac{\tan^{-1}\frac{1}{x}}{2x^2-3x+2}.dx$

Adding $2\alpha = \int_{\frac{1}{2}}^{2} \frac{\frac{\pi}{2}}{2x^2-3x+2}$

For More Join: @IITJEE_Advanced

$$\alpha=\frac{\pi}{4}\frac{1}{2}.\int_{\frac{1}{2}}^{2}\frac{1}{\left(x-\frac{3}{4}\right)^2+\left(\frac{\sqrt{7}}{4}\right)^2}$$

$$\frac{8\alpha}{\pi}=\frac{4}{\sqrt{7}}\tan^{-1}\left(\frac{x-\frac{3}{4}}{\frac{\sqrt{7}}{4}}\right)\Bigg]_{\frac{1}{2}}^{2}$$

$$\frac{8\alpha}{\pi}=\frac{4}{\sqrt{7}}\left(\tan^{-1}\frac{5}{\sqrt{7}}+\tan^{-1}\frac{1}{\sqrt{7}}\right)$$

$$\frac{8\alpha}{\pi}=\frac{4}{\sqrt{7}}\tan^{-1}\left(3\sqrt{7}\right)\Rightarrow\tan\left(\frac{2\alpha\sqrt{7}}{\pi}\right)=3\sqrt{7}$$

$$\therefore\sqrt{7}\tan\left(\frac{2\alpha\sqrt{7}}{\pi}\right)=21$$

23. Required probability $=\dfrac{\frac{2}{10}\times\frac{2}{3}}{\frac{2}{10}\times\frac{2}{3}+\frac{8}{10}\times\frac{1}{2}}=\frac{1}{4}$

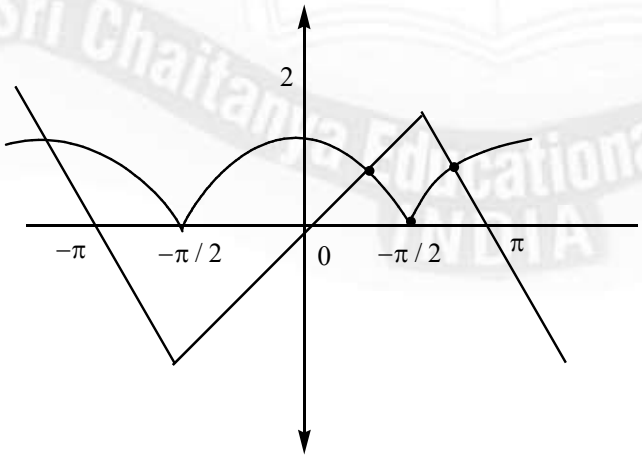
24. $N={}^{15}C_{10}2^5.(-3)$

$=2^5.3^{11}.7.11.13$

25. $\sqrt{1+\cos 2x}=\sqrt{2}\sin^{-1}(\sin x)$

$\sqrt{2}|\cos x|=\sqrt{2}\sin^{-1}(\sin x)$

$|\cos x|=\sin^{-1}(\sin x)$



$\therefore$ no.of points of intersection in $-\pi\le\pi\le\pi$ are 2 only

For More Join: @IITJEE_Advanced

PHYSICS

26. $R = \frac{A}{B}$, $A = 1, B = \frac{1}{20}$

27. 1) coefficient of viscosity (η) $(\eta) = \frac{(F/A)}{\left(\frac{dv}{dx}\right)} = \frac{ML^{-1}T^{-2}}{T^{-1}} = ML^{-1}T^{-1}$

2) Surface tension (T) $T = \frac{F}{l} = \frac{MLT^{-2}}{L} = MT^{-2}$

3) Angular momentum (L)

$L = mvr = ML^{-1}.L = ML^2T^{-1}$

4) Rotational kinetic energy (K.E)

$K.E = \frac{1}{2} \pm \omega^2 \qquad = ML^2T^{-2}$

28. for Ring $I = mR_1^2 = mK_1^2$

Radius of gyration $K_1 = R_1$

For solid sphere $I' = \frac{2}{5}m'R_2^2 = m'K_2^2$

Its radius of hydration

$K_2 = \sqrt{\frac{2}{5}}R_2$

$\therefore K_1 = K_2$

$\therefore R_1 = \sqrt{\frac{2}{5}}R_2$

$\frac{R_1}{R_2} = \sqrt{\frac{2}{5}}$

$\therefore x = 5$

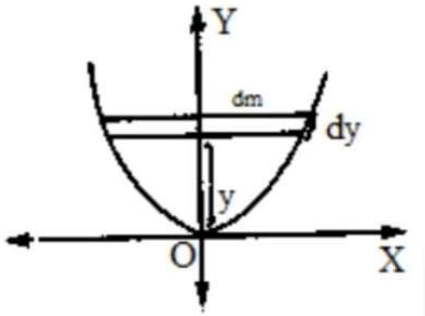
29. Given, parabolic equation

$y = kx^2 \Rightarrow x^2 = \frac{y}{k}$

Let ρ be the volume density of the paraboloid.

Consider an infinitesimal disc of radius x , width dy , mass dm at a distance y from O.

For More Join: @IITJEE_Advanced



$$\Rightarrow dm = \rho\pi x^2 dy = \pi\rho\left(\frac{y}{k}\right)dys \qquad \Rightarrow y_{COM} = \frac{\int ydm}{\int dm}$$

$$= \frac{\int_0^h \pi\rho\left(\frac{y^2}{k}dy\right)}{\int_0^h \pi\rho\left(\frac{y}{k}\right)dy} = \frac{\left[\frac{y^2}{3}\right]^h}{\left[\frac{y^2}{2}\right]_0^h} \qquad \Rightarrow y_{COM} = \frac{2h}{3}$$

30. $m_e = 27m_p \qquad V_{e,e} = 11.2 km/s$

$R_e = 3R_p \qquad V_{e,p} = ?$

$$V_e = \sqrt{\frac{2GM}{R}} \qquad \frac{V_{e,e}}{V_{e,p}} = \sqrt{\frac{M_e}{M_p}\times\frac{R_p}{R_e}} = 3$$

$$V_{e,p} = \frac{V_{e,e}}{3} = \frac{11.2}{3} = 3.7 km/s$$

31. When any input is 0 then current flows and $v_y = 0$
and when both input is 1
No current flows and $v_y = 1$

32. $$\frac{I_{max}}{I_{min}} = \left[\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right]^2 \qquad \frac{I_{max}}{I_{min}} = 9$$

33. $$K_{eq} = \frac{2k.k}{3k} + k = \frac{5k}{3}. \qquad T = \pi\sqrt{\frac{12m}{5k}}$$

34. By conservation of mechanical energy

$$U_A + K_A = U_B + K_B \qquad Mg(1) = Mg\left(\frac{1}{2}\right) + \frac{1}{2}Mv_B{}^2 \qquad v_B = \sqrt{10}\frac{m}{s}$$

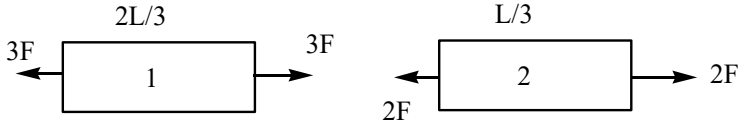
35. Total resistance R = 20 + 40 = 60 Ω
The impedance of LCR circuit is given by

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \qquad = \sqrt{60^2 + (100+20)^2} = 100$$

For More Join: @IITJEE_Advanced

Power factor, $\cos\phi = R/Z = 60/100 = 0.6$

36.



$$Y = \frac{F.L}{A\Delta L}, \Delta L = \frac{FL}{AY}$$

Elongation in rod of length $\frac{2L}{3}$ $\qquad \Delta L_1 = \frac{3F \times \frac{2L}{3}}{AY} = \frac{2FL}{AY}$

Elongation in rod of length $\frac{L}{3}$

$$(\Delta L)_2 = \frac{2F\frac{L}{3}}{AY} = \frac{2FL}{3AY}$$

Net elongation $(\Delta L)_{net} = (\Delta L)_1 + (\Delta L)_2$

$$= \frac{2FL}{AY} + \frac{2FL}{3AY} = \frac{8FL}{3AY}$$

37. $R = \frac{\rho l}{A} = \frac{\rho V}{A^2} \qquad \therefore \frac{R_A}{R_B} = \frac{A_B{}^2}{A_A{}^2} = \frac{r_B{}^4}{r_A{}^4} \qquad = R_A = 32\Omega$

38. $x = 13 - 7 = 6m$

According to conservation of energy

$$k_i + u_i = k_f + u_f, \frac{1}{2}400 \times 36 = \frac{1}{2}mv^2, v = 60m/s$$

$$N = \frac{mv^2}{r} = 2880N$$

39. $\lambda = \frac{h}{p} \qquad \lambda = \frac{h}{\sqrt{2mk}}$

$k \propto \frac{1}{\lambda^2} \qquad \frac{k_2}{k_1} = \left(\frac{\lambda_1}{\lambda_2}\right)^2$

$\lambda_2 = \frac{175}{100} \quad \lambda_{1,2} = \frac{16}{49}k_1$

40. $m = 2 = \frac{-v}{u}$

$u = 5cm$

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \quad = \frac{1}{10} + \frac{1}{(-5)} = \frac{1-2}{10} = \frac{-1}{10} \quad f = -10cm$

41. Intensity of light $I = \frac{nhv}{A}$

For More Join: @IITJEE_Advanced

Here n is no.of photons per unit time

$n=\frac{1A}{hv}$ so on increasing frequency υ, n decrease taking intensity constant

$k_{\max}=hv-\phi$

So on increasing v, kinetic energy increases

42. Conceptual

43. Translational degree of freedom = 3 rotational degree of freedom = 2

44. Conceptual

45. B.E per nucleon is maximum for Fe^{56}

46. $C_1=\frac{(A/3)\varepsilon_0}{d}, C_2=\frac{(A/3)(\varepsilon_0)}{2d}$

$C_3=\frac{(A/3)\varepsilon_0}{3d}$ $C_{net}=C_1+C_2+C_3$

$=\frac{(A/3)\varepsilon_0}{d}+\frac{(A/3)(\varepsilon_0)}{2d}+\frac{(A/3)\varepsilon_0}{3d}$ $=\frac{(A/3)\varepsilon_0}{d}\left[1+\frac{1}{2}+\frac{1}{3}\right]=\frac{11A\varepsilon_0}{6\times 3d}$

47. $\phi=\bar{E}.\bar{A}=\left(2\hat{i}+4\hat{j}+6\hat{k}\right)\times 10^3.A\hat{j}$, $A=15cm^2$

48. Area of circle, $W=\pi\left(\frac{\Delta V}{2}\right).\left(\frac{\Delta P}{2}\right)$

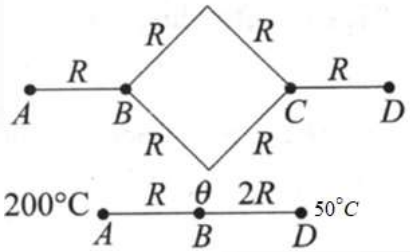
$W=\frac{\pi}{4}(300-100)\times(300-100)=\frac{200\times 200\times 10^{-3}\times\pi}{4}=314\times 10^{-1}$ Joule

49. $t=\frac{2l}{v}$

Induced emf $e=Bvl$

Heat produced $H=\frac{e^2t}{R}=\frac{(Bvl)^2}{R}\frac{2l}{v}=\frac{2B^2vl^3}{R}$

50.



Let the thermal resistance of each rod be R

The two resistances connected along two paths from B to C care equivalent to 2R each and their parallel combination is R

Effective thermal resistance between B and D = 2R

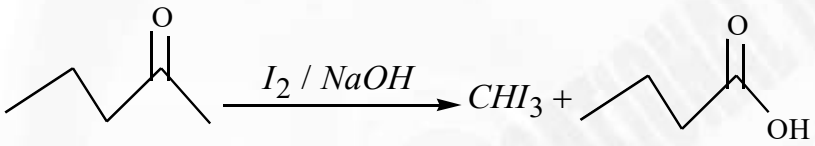
For More Join: @IITJEE_Advanced

$$\frac{200-50}{3R}=\frac{200-\theta_B}{R}$$

$$\frac{150}{3R}=\frac{200-\theta_B}{R}$$

$$\theta_B=150^{\circ}C$$

CHEMISTRY

51. 

52. All are Isoelectronic

53. Both assertion and reason are true and reason is the correct explanation of assertion. Isotopes have the same atomic number. i.e. same number of electrons which are responsible for their chemical behaviour

54. Assertion is false but Reason is true
Combustion of 16 g of methane gives 36 g of water
$CH_4+2O_2\rightarrow CO_2+2H_2O$

55. In octahedral complexes, when $\Delta_o<P$ high spin complexes are formed. When $\Delta_o>P$ low spin complexes are formed.in tetrahedral complexes because of $\Delta_t<P$, low spin complexes are rarely formed. Mostly high spin complexes are formed.

56. ICL- linear , ICl_3-T-shape, ClF_5-Square pyramidal , IF_7-Pentagonal bipyramidal

57. $22.4\rightarrow 1$

$$0.224\rightarrow\frac{0.224}{22.4}=\frac{1}{100}$$

$$\text{w}\,CaCO_3=n\times M.w=\frac{1}{100}\times 100=1g$$

58. Conceptual

59. Stability of +5 oxidation state decreases top to bottom and +3 oxidation state increases top to bottom due to inert pair effect. Mean while compound having +5 oxidation state of Bi is BiF_5.it is due to small size and high electronegativity of fluorine.

60. $\Delta_t=\frac{4}{9}\Delta_0$ $\Delta t=\frac{4}{9}\times 18000=8000cm^{-1}$

61. Wolf-kishner reduction

62. $t_{75}=2t_{50}$ $t_{87.5\%}=3t_{50\%}$

For More Join: @IITJEE_Advanced

$\therefore t_1 = 2t_{50}$ $t_2 = 3t_{50}$ $\frac{t_1}{t_2} = \frac{2}{3}$

63. $p_{(g)} \rightleftarrows Q_{(g)} + R_{(g)}$

$t = 0$ a moles

t_{eq} $(a - a\alpha)$ $(a\alpha)$ $(a\alpha)$

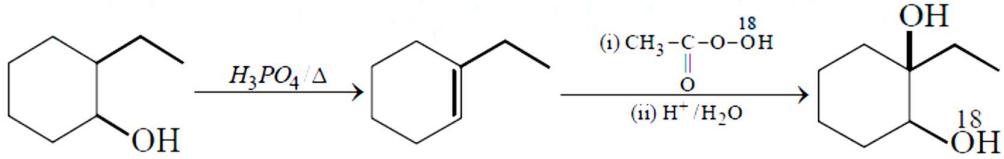
If $P_{Total} = P_1$ Then $K_{P_1} = \dfrac{\left(\frac{(\alpha)}{1+\alpha}P_1\right)\left(\frac{\alpha}{1+\alpha}P_1\right)}{\left(\frac{1-\alpha}{1+\alpha}P_1\right)}$

$A_{(g)} \rightleftarrows 2B_{(g)}$

$t = 0$ b moles

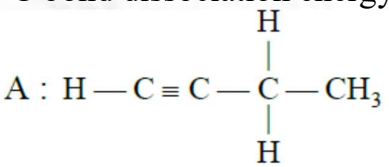
t_{eq} $(b - b\alpha)$ $(2b\alpha)$

$K_{p2} = \dfrac{\left(\frac{2\alpha}{1+\alpha}p_2\right)^2}{\left(\frac{1-\alpha}{1+\alpha}p_2\right)}$ Given $K_{P_1} / K_{p2} = \frac{1}{9}$

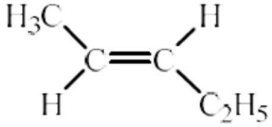
64. 

65. $\left[H^+\right]_{final} = \dfrac{1 \times 0.01}{(1+999)} = 10^{-5}$ $P^H = 5$

66. C-C bond dissociation energy is 355 kJ /mol which is highest amongst the carbon family.

A : 

B : $CH_3 - C \equiv C - CH_2 - CH_3$

C : 

67.

68. $Ca(NO_3)_2 \rightleftharpoons Ca^{+2} + 2NO_3^-$

$Initial$ 1 0 0

$Final$ $1-\alpha$ α 2α

Total no of moles at equilibrium = $1 - \alpha + \alpha + 2\alpha = 1 + 2\alpha$

For More Join: @IITJEE_Advanced

$i = 1 + 2\alpha = 1 + 2(0.7) = 2.4$ $\quad \frac{P^\circ - P_s}{P^\circ} = x_A = \frac{n_A}{n_A + n_B}$

$P_\circ = 760, n_A = \frac{W_A}{mW_A} = \frac{7}{16C_4}, n_B = \frac{W_B}{M.W_B} = \frac{100}{18}$

$\frac{M_{obs}}{M_{cal}} = 1 + 2\alpha \Rightarrow M_{calculate} = \frac{164}{2.4} = 68.33$

$n_A = \frac{7}{68.33}$ $\quad \frac{P_\circ - P_S}{P_S} = \frac{7 \times 18}{68.33 \times 100} = 0.0184$ $\quad P_S = \frac{760}{1.0184} = 746.26 mmHg$.

69. $E_{Cu^{2+}|Cu} = E^\circ_{Cu^{2+}|Cu} - \frac{0.06}{2}.\log\frac{1}{[Cu^{2+}]}$

$= E^\circ_{Cu^{2+}|Cu} + 0.03\log[Cu^{2+}] = 0.34 + 0.03 \times \log(0.1) = 0.31V$ $\quad \therefore E_{Cu/Cu^{2+}} = -0.31V$

70. order $= \frac{3}{2} - 1 = \frac{1}{2}$

71. 1st IP of B=801 Kg/mol
1st IP of Be=1757 Electron efficiency: - S>0

72. $P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$

$x + y = 6$

73. $4H^+ + MnO_4^- \rightleftarrows MnO_2 + 2H_2O$

$\Delta G^0 = -3F\left(\frac{1.5 \times 5 - 2 \times 1.25}{3}\right) = -5F$ X=5

74. $C_2ClFBrH$ having 6 isomers

75. $W = 0.50g$
Back titration un used acid

$N_1 = 1N$ NaOH $\quad \%N = \frac{1.4 \times 1 \times 20}{0.5}$

$V_1 = ?$ $\quad N_2 = 0.5$ $\quad = 14 \times 4 = 56$

$V_2 = 60$

From $N_1V_1 = N_2V_2$ $\quad \Rightarrow V_1 = \frac{0.5 \times 60}{1} = 30ml$

$\therefore$ Used acid= 50-30=20 ml

For More Join: @IITJEE_Advanced